Title of the project: -

DESIGN AND APPLY CONTEXT-AWARE ACCESS CONTROLS BASED ON USER IDENTITY

Overview: -

- **Implement context-aware access control by dynamically granting or restricting access based on user identity, location, device, and risk level at the time of the request.**

List of teammates–

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List of Vulnerability Table:-

S.no	Vulnerability Name	CWE/CVE - No
1	Lack of Context-Aware Access Control Based on User Identity	CWE-284
2	Improper Authentication	CWE-287
3	Missing Authorization Check	CWE-862

REPORT: -

Vulnerability Name: Lack of Context-Aware Access Control Based on User Identity

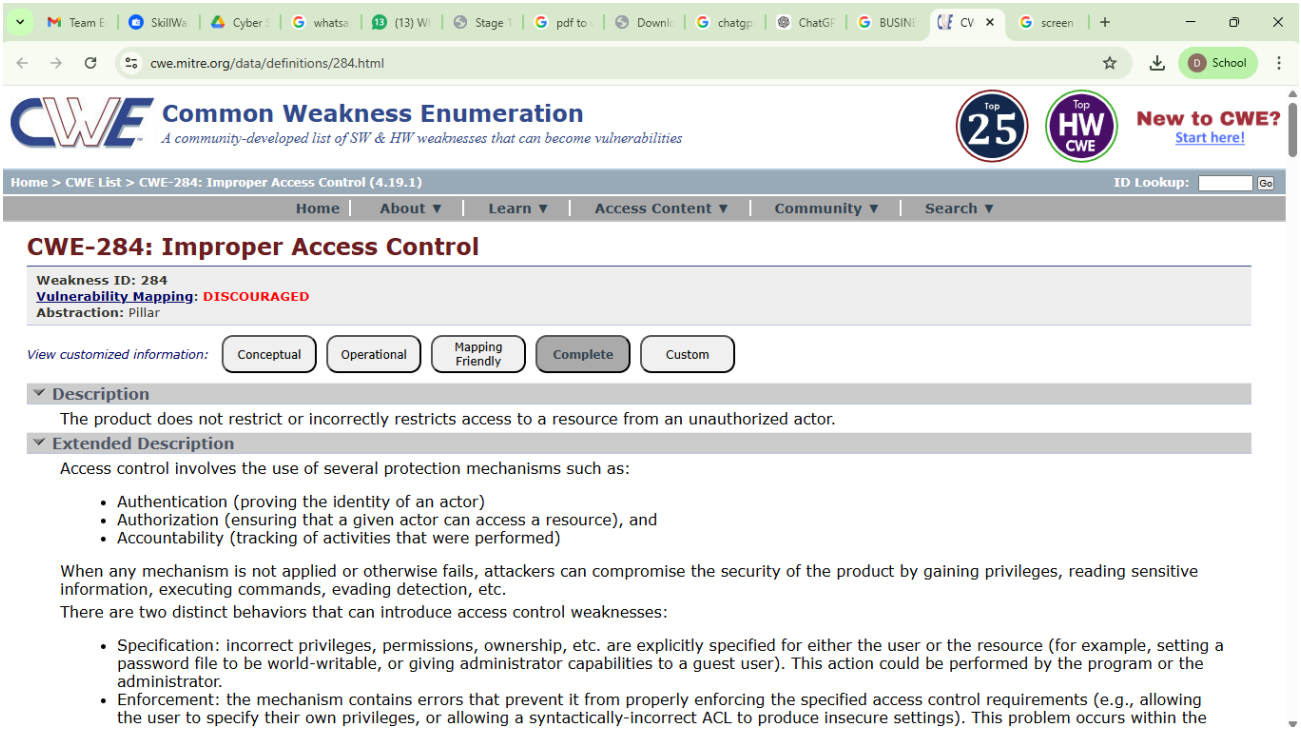
CWE/CVE: CWE-284

OWASP/SANS Category: A01:2025 Broken Access Control

Description:

Lack of context-aware access control occurs when a system grants access to resources without properly verifying the user's identity and contextual attributes such as device, location, session behavior, or risk level. In such cases, access decisions rely only on basic authentication and do not consider additional security factors that help validate whether the access request is legitimate. In a secure architecture such as the Zero Trust model, access should always be verified based on multiple contextual parameters. If these controls are missing, attackers or unauthorized users may gain access to sensitive systems or data by using compromised credentials or unauthorized devices. This can lead to privilege misuse, data exposure, and potential compromise of critical resources.

Business Impact: -



cwe.mitre.org/data/definitions/284.html

Common Weakness Enumeration

A community-developed list of SW & HW weaknesses that can become vulnerabilities

Top 25 | Top HW CWE | New to CWE? Start here!

Home > CWE List > CWE-284: Improper Access Control (4.19.1) | ID Lookup: Go

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CWE-284: Improper Access Control

Weakness ID: 284
Vulnerability Mapping: DISCOURAGED
Abstraction: Pillar

View customized information: Conceptual | Operational | Mapping Friendly | Complete | Custom

Description

The product does not restrict or incorrectly restricts access to a resource from an unauthorized actor.

Extended Description

Access control involves the use of several protection mechanisms such as:

- Authentication (proving the identity of an actor)
- Authorization (ensuring that a given actor can access a resource), and
- Accountability (tracking of activities that were performed)

When any mechanism is not applied or otherwise fails, attackers can compromise the security of the product by gaining privileges, reading sensitive information, executing commands, evading detection, etc.

There are two distinct behaviors that can introduce access control weaknesses:

- Specification: incorrect privileges, permissions, ownership, etc. are explicitly specified for either the user or the resource (for example, setting a password file to be world-writable, or giving administrator capabilities to a guest user). This action could be performed by the program or the administrator.
- Enforcement: the mechanism contains errors that prevent it from properly enforcing the specified access control requirements (e.g., allowing the user to specify their own privileges, or allowing a syntactically-incorrect ACL to produce insecure settings). This problem occurs within the